# SUPPLEMENTARY APPENDIX

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## Supplementary Table – S1: Search criteria

**PubMed search criteria - 48**

(cardiopulmonary resuscitation[Title/Abstract] OR cardiac arrest[Title/Abstract] OR chest compression[Title/Abstract] OR life support[Title/Abstract] OR CPR) AND (personal protective equipment[Title/Abstract] OR PPE[Title/Abstract] OR N95 mask[Title/Abstract] OR CBRN[Title/Abstract])

**EMBASE search criteria – 52**

('resuscitation':ab,ti OR 'heart arrest':ab,ti OR 'chest compression depth':ab,ti OR 'chest compression rate':ab,ti OR 'chest compression fraction':ab,ti) AND ('protective equipment':ab,ti OR 'n95 mask':ab,ti)

**Web of Science search criteria – 4**

| | | |
|---|---|---|
| # 3 | 4 | #2 AND #1<br>*Indexes=SCI-EXPANDED Timespan=All years* |
| # 2 | 1,023 | TI=(personal protective equipment OR PPE OR N95 mask OR CBRN)<br>*Indexes=SCI-EXPANDED Timespan=All years* |
| # 1 | 8,160 | TI=(cardiopulmonary resuscitation OR chest compression OR life saving procedures)<br>*Indexes=SCI-EXPANDED Timespan=All years* |

# Supplementary Table – S2: PRISMA Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 1 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | 3 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 3 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | PROSPERO (CRD42020192031) |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 4 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 4 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | Available in Supplement |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 4 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 5 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 5 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how | 5 |

| | | | |
|---|---|---|---|
| | | this information is to be used in any data synthesis. | |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 5, 6 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 5, 6 |
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 5 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 6 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 7, Figure - 1 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 7, 8 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 8 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 8, 9, Figure – 2 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 8, 9, Figure – 2 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 8, Table – S1 (Supplement) |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 9, Figure – S1, S2, S3 and S4 (Supplement) |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 10, 11, 12 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 13 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 14 |
| **FUNDING** | | | |

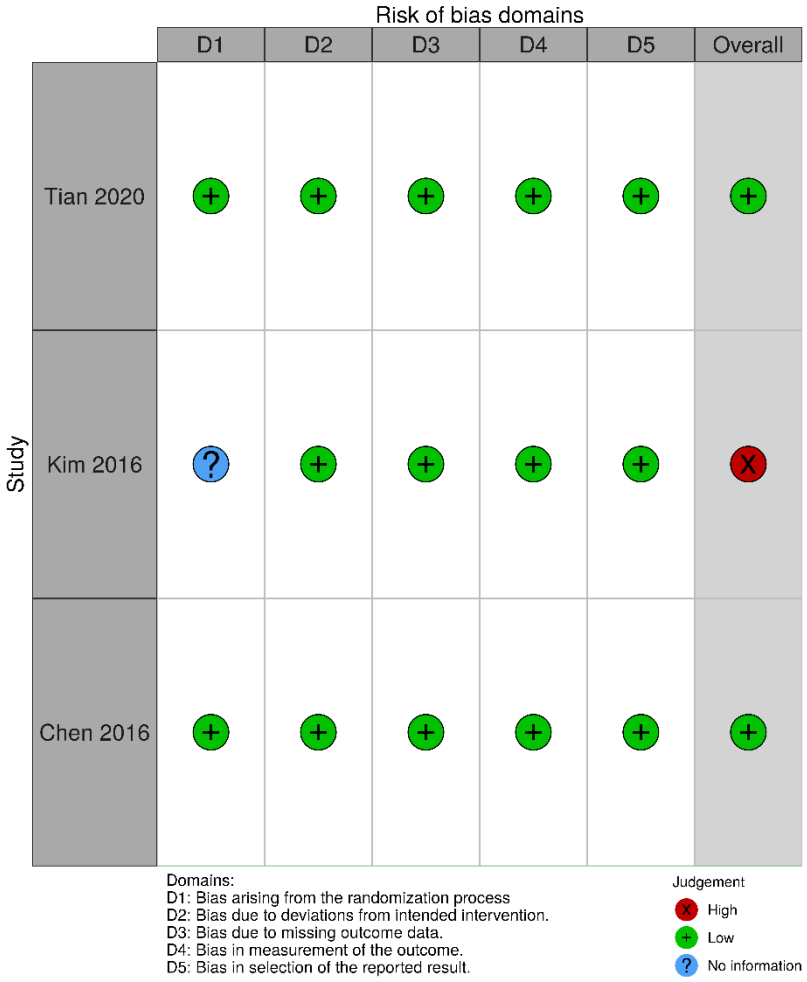
| | | | |
|---|---|---|---|
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 22 |

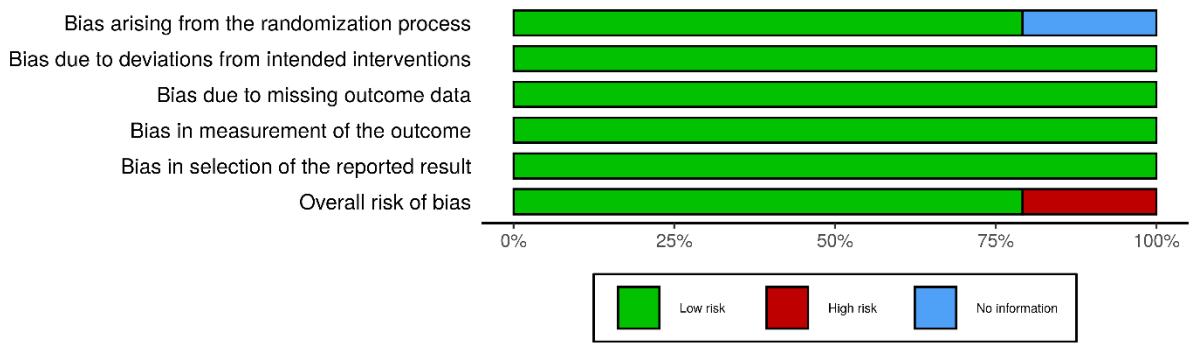
*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

## Supplementary Table – S3: Table showing studies that had provided information about proportion of adequate chest compression rate, depth and recoil

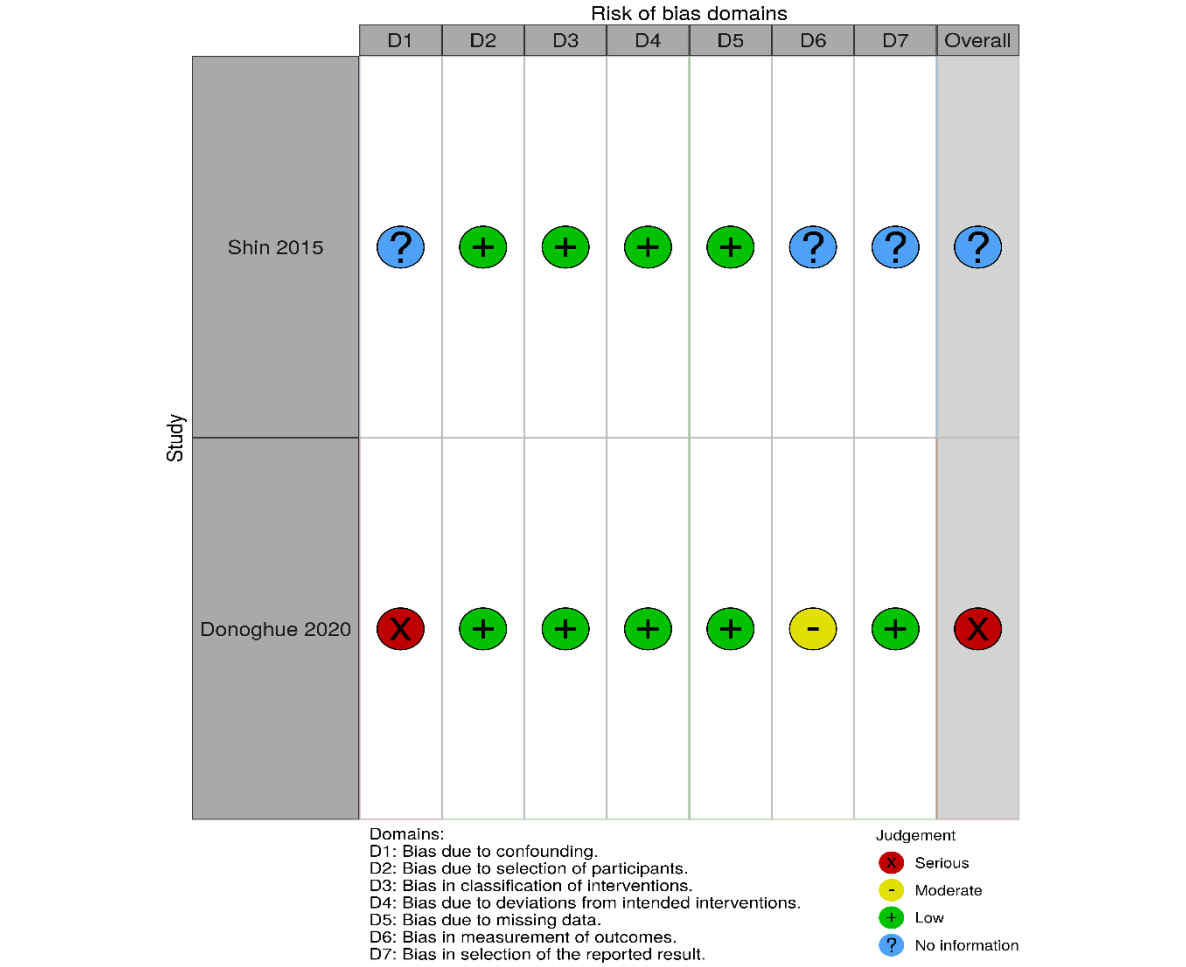
| Study ID | Sample size | | Proportion of adequate chest compression rate provided (%) | | Proportion of adequate chest compression depth provided (%) | | Proportion of adequate chest recoil allowed (%) | |
|---|---|---|---|---|---|---|---|---|
| | 'no-PPE' arm | 'PPE' arm | 'no-PPE' arm | 'PPE' arm | 'no-PPE' arm | 'PPE' arm | 'no-PPE' arm | 'PPE' arm |
| Chen 2016 | 40 | 40 | 80.7 + 15.5 | 67.7 + 18.9 | 67.5 + 15.6 | 41.3 + 17.1 | | |
| Kim 2016 | 20 | 20 | 62.4 + 34.7 | 49.6 + 37.6 | 55.5 + 40.1 | 57.1 + 43.6 | | |
| Tian 2020 | 40 | 40 | 75 + 19 | 61 + 19 | 90 + 14 | 67 + 16 | 98 + 5 | 91 + 16 |

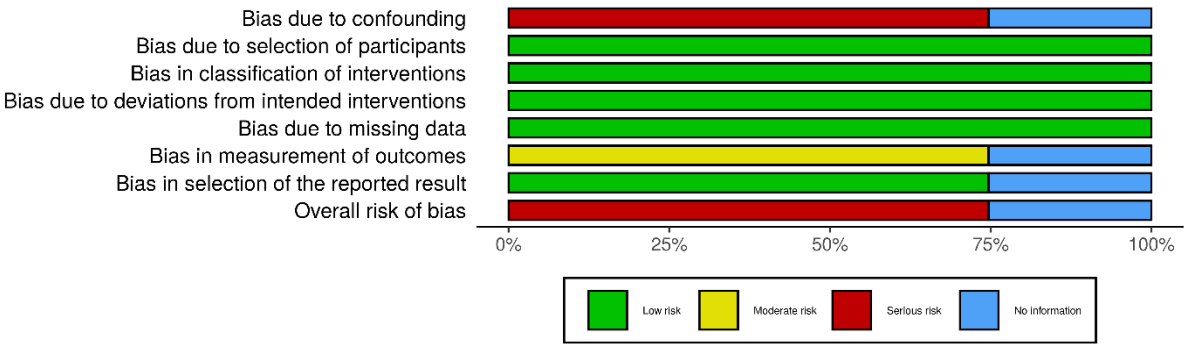
# Supplementary Figure – S1: Risk of Bias analysis for randomised trials using Cochrane ROB-2 tool



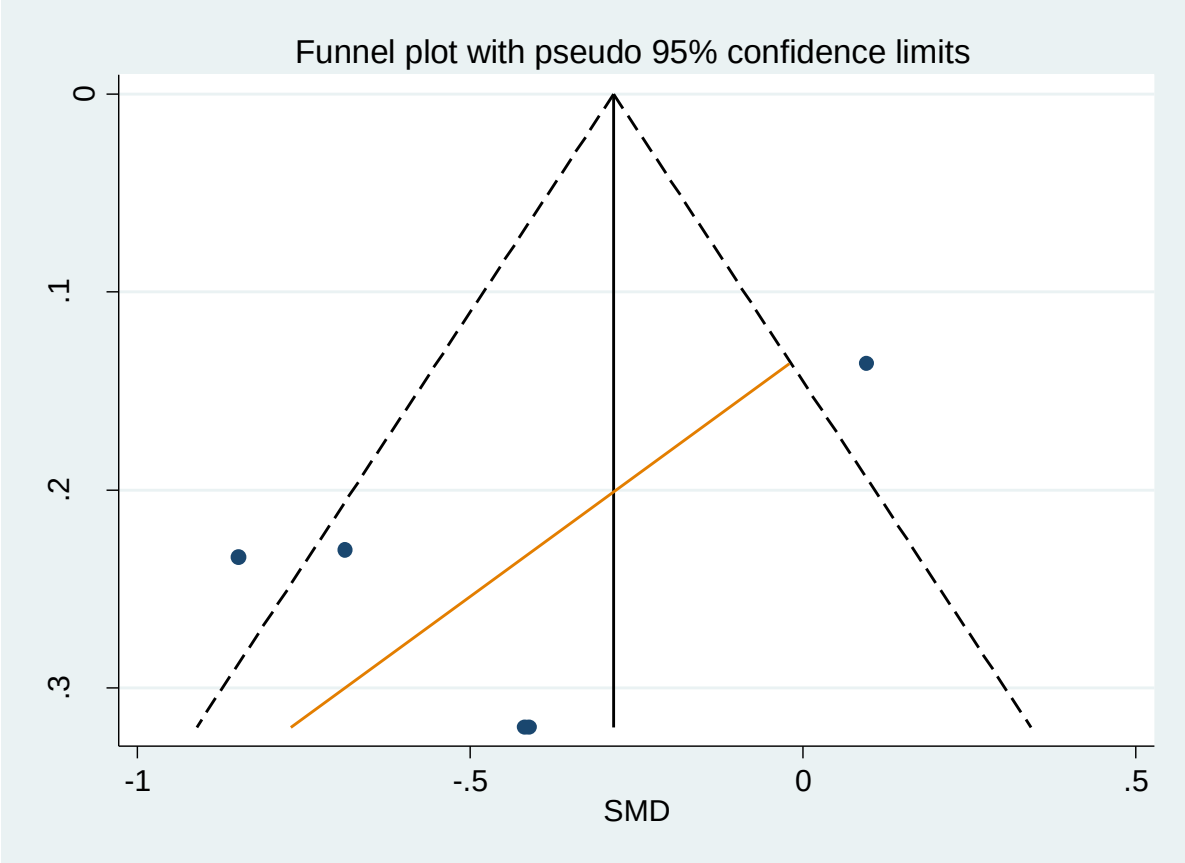



# Supplementary Figure – S2: Risk of Bias analysis for non-randomised studies using ROBINS-I tool (Risk Of Bias In Non-randomized Studies of Interventions)



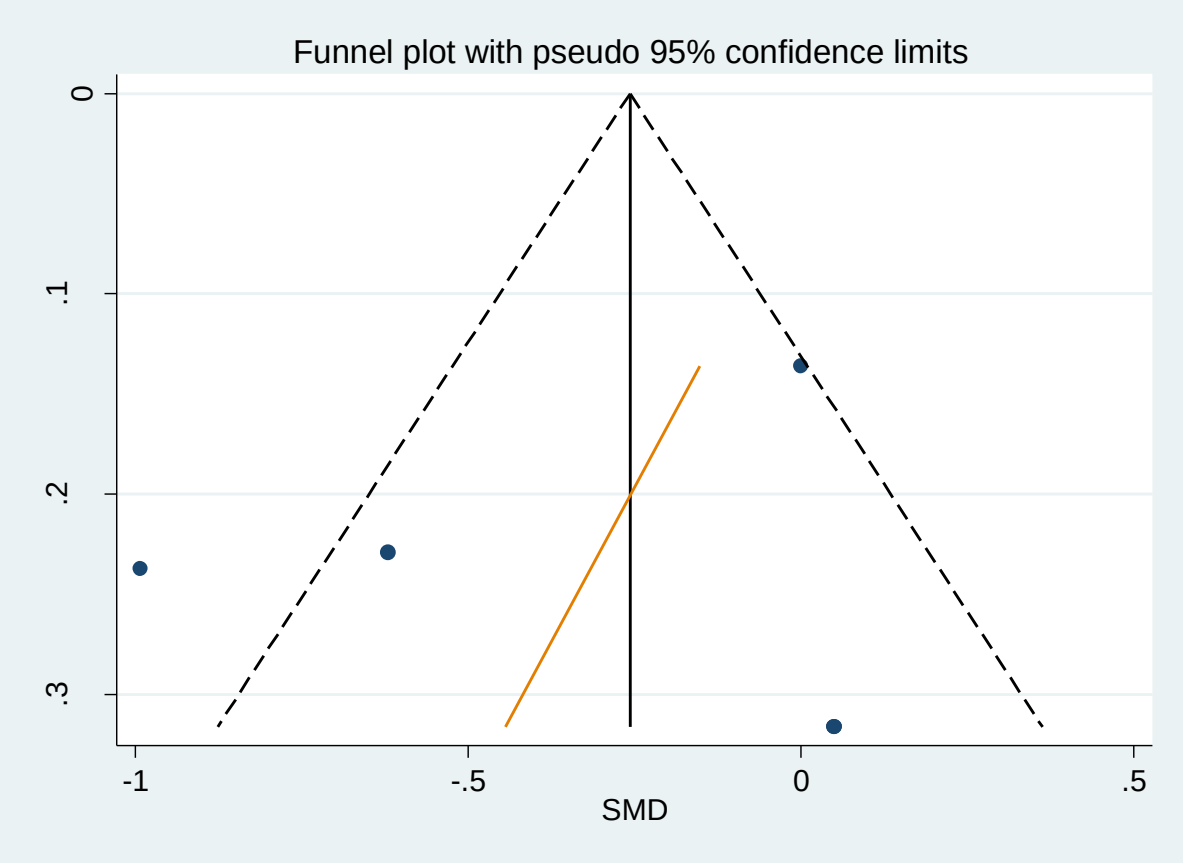



## Supplementary Figure – S3: Funnel plot for publication bias among studies that provided data on chest compression rate




* Red line depicts the Egger's regression line with Egger's regression coefficient – 0.166

## Supplementary Figure – S4: Funnel plot for publication bias among studies that provided data on chest compression depth




* Red line depicts the Egger's regression line with Egger's regression coefficient – 0.646